US Census Income and Poverty Analysis (2015)

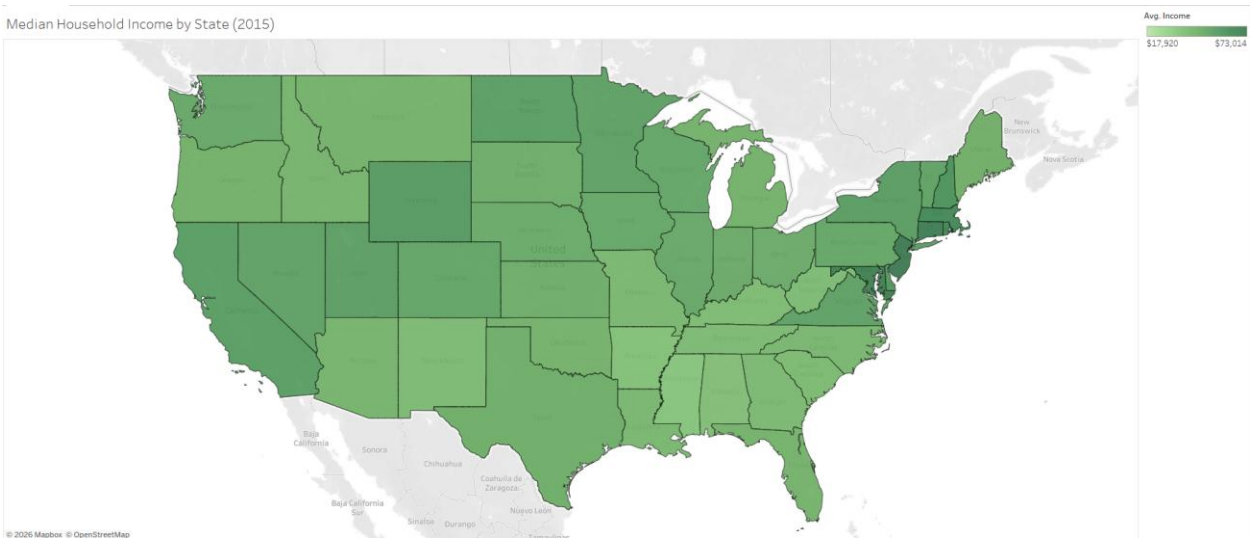
By Matt Beckmann

This project analyzes economic and demographic data patterns collected at the county and state level from across the United States using the 2015 US Census Demographic Data set from Kaggle. Tableau was used to create several interactive visualizations exploring relationships between median household income, poverty rates, and geographic trends across states and counties. The purpose of the analysis was to identify regional economic differences and examine how poverty and household income levels are related throughout the United States.

Visualization 1 – Median Household Income by State (2015)

Tableau Public Link

Income Map



Summary of Insight

This visualization displays the average median household income (measured in US dollars) for each state in the United States during 2015 using a filled geographic map. Darker shades of green indicate states with higher median household incomes, while lighter shades indicate lower incomes.

The visualization suggests that household income levels vary significantly across different regions of the US. Northeastern states and several western states generally displayed higher median household incomes, while many southern states appeared to have lower average income levels. The geographic clustering visible on the map highlights broader regional economic differences across the country.

Design Choice

A filled map was chosen for this visualization because the data contains a strong geographic component. Using a map allows regional patterns and economic differences between states to be identified quickly and intuitively. A green color gradient was selected because of its common association with the color of money and wealth, while also providing a clear visual distinction between higher and lower income levels that remains easy to interpret. Darker shades naturally draw attention to states with higher incomes, making the visualization more readable.

Resources

- Kaggle US Census Demographic Data: https://www.kaggle.com/datasets/muonneutrino/us-census-demographic-data/data
- Tableau Documentation: https://www.tableau.com/support/help

Visualization 2 – Poverty Rate vs Median Household Income

Tableau Public Link

Poverty vs Income Scatterplot



Summary of Insight

This visualization uses a scatterplot to examine the relationship between county poverty rates and median household income levels across the US. Each point represents an individual county. Poverty rate is measured as a percentage, while median household income is measured in US dollars.

The visualization shows a strong negative relationship between poverty and median household income. Counties with higher poverty rates generally tend to have lower household income levels. The logarithmic trend line further emphasizes this relationship and suggests that increases in poverty rates are associated with substantial decreases in household income, particularly at lower poverty levels.

This insight is relevant because it highlights the economic disparities that exist between counties throughout the United States and demonstrates how closely poverty and income levels are connected.

Design Choice

A scatterplot was selected because it is one of the most effective visualization types for identifying relationships and correlations between two continuous variables. Poverty rate and median household income are both numerical measures, making a scatterplot an appropriate choice for examining trends and clustering patterns within the data.

A logarithmic trendline was used instead of a linear trendline because it better matched the curved relationship visible in the data. Green was selected as the primary color to maintain visual consistency throughout the project while remaining readable with a large number of data points and providing good contrast against the trendline.

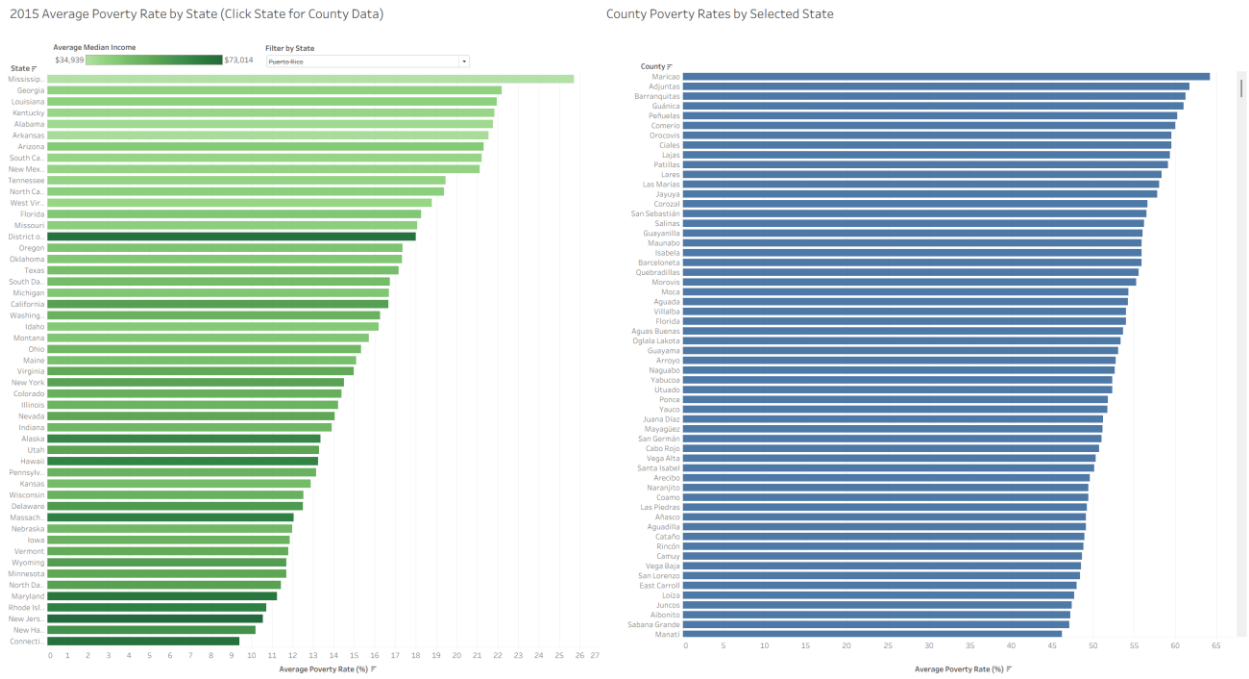
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Visualization 3 – Average Poverty Rate by State Dashboard

Tableau Public Link

Dashboard



Summary of Insight

This dashboard displays average poverty rates by state and allows users to interactively explore county-level poverty rates by selecting a state from the chart. States are sorted from highest to lowest poverty rate, while the second chart dynamically updates to display poverty rates for counties within the user's selected state. Bar coloring on the state chart represents average median household income levels.

The dashboard demonstrates that poverty rates vary substantially across both states and counties. States with higher statewide poverty rates, such as Mississippi and Louisiana, also contain many counties with elevated poverty levels. The interactive county comparison further highlights how poverty distribution can vary within individual states, reinforcing the broader relationship between poverty and income levels observed throughout the analysis.

Design Choice

A dashboard format was selected to provide an interactive component and allow users to explore both state-level and county-level poverty patterns within a single visualization. Horizontal bar charts were used because they allow large numbers of categories to be compared and ranked clearly from highest to lowest values.

A green gradient color palette was chosen for the state chart because it provides visual contrast between higher and lower median household income levels while maintaining consistency with the overall visual style of the project. The county chart uses a neutral blue color palette to visually separate county-level data from the state-level comparison and improve readability.

Resources

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- Tableau Documentation: https://www.tableau.com/support/help

Conclusion

The visualizations in this project identified clear economic differences across states and counties within the United States. Higher poverty rates were generally associated with lower median household income levels, while substantial variation also existed within individual states at the county level. The dashboard interaction improved the ability to explore these patterns more closely and provided additional context beyond statewide averages alone.